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### HINGING DRYWALL APPARATUS AND METHOD

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#### Abstract

A drywall system is provided including a first drywall panel and a second drywall panel connected to the first drywall panel via a main hinge. A first drywall flap connected to the first drywall panel via a first hinge, and a second drywall flap connected to the second drywall panel via a second hinge. The first drywall panel, the second drywall panel, the first drywall flap and the second drywall flap can hinge to assume at least one corner configuration in which the first and second flaps lie on a first plane, the first drywall panel lies on a second plane and the second drywall panel lies on a third plane, such that the first plane, the second plane and the third plane are mutually orthogonal. A coating can be provided on an external surface of the framing drywall liner for strengthening the groove which, in embodiments can be polyester strip that extends over portions of the first and second gypsum-containing segments on either side of the of the groove.

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## Background/Summary

CROSS-REFERENCE TO RELATED APPLICATION(S) [0001] This application is a continuation of U.S. patent application Ser. No. 17/992,000 filed Nov. 22, 2022, which is a continuation-in-part of U.S. patent application Ser. No. 17/134,233, filed Dec. 25, 2020, which is a continuation of U.S. patent application Ser. No. 16/333,337, filed Mar. 14, 2019, which is a 371 continuation of PCT/CA2017/000204, filed Sep. 14, 2017, which claims priority to U.S. provisional application No. 62/395,161, filed Sep. 15, 2016, the contents of which are incorporated herein by reference herein.

### FIELD OF THE INVENTION

[0002] This invention relates to a drywall apparatus and method, and more particularly to a hingeable drywall apparatus and method of making same.

### BACKGROUND OF THE INVENTION

[0003] Drywall or gypsum boards are used in modern construction as a fire resistant smooth cladding surface for interior walls or ceilings. A drywall board is made of gypsum material sandwiched between two sheets of drywall paper or liner. In a typical manufacturing process, drywall board is formed by mixing calcium sulphate hemihydrate (known as stucco) with water and other additives to produce a slurry, which is deposited between two parallel sheets of drywall paper that form an envelope. The envelope is extruded through an orifice. The extrusion forms a continuous ribbon, several hundred feet in length, of a gypsum slurry core that is enclosed by the two sheets of drywall paper. The parallel sheets of paper are provided from a roll that continuously unwinds to supply the board line. The two sheets of drywall paper are typically glued together near the edges of the board. The ribbon is cut into individual boards. A board kiln completes the drying process.

[0004] In the construction of buildings, drywall boards are commonly used to build interior walls and corners. The edges of drywall boards are often tapered such that where two drywall boards abut, a cove or depression is formed. The cove is first filled with joint compound and then tape is pressed into the joint compound along the full length of the cove. More joint compound is then placed over the tape before the first sanding of the resulting joint is performed. Iterations of joint compound application and sanding are performed as needed.

[0005] The area where two boards abut at a corner is often more difficult to finish than where two boards abut along a flat portion of a wall or ceiling. At an inner (less than 180 degrees) or outer (greater than 180 degrees) corner, taping, joint compound application and sanding are more cumbersome. The joint application and sanding process is usually performed several times, even by an experienced and highly skilled drywall finisher, before the corner joint takes on the appearance of a cleanly, integrally formed corner area with no visually perceptible joint areas. The finishing process is especially time consuming and highly dependent upon the skill of the drywall finisher.

As will be appreciated, this adds to the overall cost of constructing any structure where drywall is used and increases the time needed for drywall finishing.

[0006] The above finishing process can be particularly troublesome for home remodelling applications undertaken by “do-it-yourself” persons who do not have extensive experience in working with drywall finishing and have not acquired the necessary skill to finish inner and outer corner areas of a structure in a manner that produces clean, well-finished corner areas free from visual imperfections. Whereas the portions of adjacent drywall boards having tapered edges that meet along a flat wall or ceiling can usually be finished adequately by even a “do-it-yourself” person, the inner and outer corner areas are usually difficult and time consuming for such persons to finish.

[0007] When forming outer corners between two drywall boards, it has previously been necessary to nail or screw a metal corner section over the corner before taping and applying joint compound to the corner. The metal corner member has to be attached carefully such that it forms a straight vertical edge. If this component is not attached properly, a “wavy”, non-linear edge will be formed, requiring even further finishing efforts.

[0008] It is therefore a principal object of the present invention to provide an apparatus and method for enabling inner and outer drywall corners to be quickly and easily constructed.

[0009] It is still a further object of the present invention to provide an apparatus and method which is inexpensive to produce, easy to ship and install, and which further does not add appreciably to the overall construction costs when working with drywall boards, and which further enables the drywall finishing process to be performed with reduced labour time and skill level.

[0010] It is still another object of the present invention to provide an apparatus and method which can be readily adapted for forming either an inner corner or an outer corner area.

#### SUMMARY OF THE INVENTION

[0011] Described herein is a drywall apparatus including drywall liner having an external side and an internal side opposite the external side. A first gypsum-containing segment is disposed on the internal side of the drywall liner, and a second gypsum-containing segment disposed on the internal side of the drywall liner such that a groove exists between the first segment and the second segment. The drywall liner connects the first segment to the second segment such that the first segment and the second segment are capable of hinging about a hinge axis parallel to the groove. The first segment and the second segment are also capable of assuming a flat position in which the first segment and second segment are coplanar, such that when in the flat position, the drywall liner is capable of having a width at the groove that is at least 40 thousands of an inch. The drywall apparatus further includes a non-metallic strengthening material in at least part of the groove for reinforcement.

[0012] Also described herein is a method of constructing a hingeable drywall apparatus including the steps of providing finishing drywall liner having an internal side and an external side and then pouring gypsum slurry onto the internal side of the finishing drywall liner. The method also includes the steps of forming a first groove on a framing drywall liner, the first groove having an appropriate shape and placing the framing drywall liner on the gypsum slurry so that the gypsum slurry is sandwiched between the finishing drywall liner and the framing drywall liner. The method further includes allowing the gypsum slurry to set to a hardened gypsum material, wherein, during the step of allowing and before the gypsum slurry has completely set to the hardened gypsum material, the shape of the first groove is impressed upon the slurry resulting in a second groove in the hardened gypsum material that is co-linear and mates with the first groove to thereby form a hinge at the first and second grooves.

[0013] Also described herein is a drywall system including a first drywall panel and a second drywall panel connected to the first drywall panel via a main hinge. A first drywall flap connected to the first drywall panel via a first hinge, and a second drywall flap connected to the second drywall panel via a second hinge. The first drywall panel, the second drywall panel, the first

drywall flap and the second drywall flap can hinge to assume at least one corner configuration in which the first and second flaps lie on a first plane, the first drywall panel lies on a second plane and the second drywall panel lies on a third plane, such that the first plane, the second plane and the third plane are mutually orthogonal.

[0014] Also described herein is a coating on an external surface of the framing drywall liner for strengthening the groove which, in embodiments can be polyester strip, such as BoPET (Biaxially-oriented polyethylene terephthalate), that extends over portions of the first and second gypsum-containing segments on either side of the of the groove.

[0015] In a further aspect, a drywall apparatus is described, comprising a finishing drywall liner; a framing drywall liner; a first gypsum-containing segment between the finishing drywall liner and the framing drywall liner; and a second gypsum-containing segment between the finishing drywall liner and the framing drywall liner, said first gypsum-containing segment and second gypsum-containing segment connected by a hinge formed from the finishing drywall liner and the framing drywall liner to allow an angle between the first gypsum-containing segment and the second gypsum-containing segment to vary, such that the hinge allows the first gypsum-containing segment and the second gypsum-containing segment to assume a flat position in which the first gypsum-containing segment and the second gypsum-containing segment are coplanar, wherein a) in the flat position, the framing drywall liner includes a groove having a longitudinal length running along the hinge, b) the hinge allows the first gypsum-containing segment and the second gypsum-containing segment to assume a corner position in which the first gypsum-containing segment and the second gypsum-containing segment form an angle therebetween of 90 degrees and c) the framing drywall liner is integral across the groove for most of the longitudinal length, further comprising: a first flap hingeably connected to the first gypsum-containing segment; and a second flap hingeably connected to the second gypsum-containing segment, wherein the apparatus can assume a corner configuration in which the first flap and the second flap lie on one plane, the first gypsum-containing segment lies on a second plane and the second gypsum-containing segment lies on a third plane such that the first plane, the second plane and the third plane are mutually orthogonal.

[0016] The above aspects can be attained by a drywall apparatus comprising a finishing drywall liner; a framing drywall liner; a first gypsum-containing segment between the finishing drywall liner and the framing drywall liner; and a second gypsum-containing segment between the finishing drywall liner and the framing drywall liner, said first segment and second segment connected by a hinge formed from the finishing drywall liner and the framing drywall liner to allow an angle between the first segment and the second segment to vary, such that the hinge allows the first segment and the second segment to assume a flat position in which the first segment and the second segment are coplanar, wherein a) in the flat position, the framing drywall liner includes a groove having a longitudinal length running along the hinge, b) the hinge allows the first segment and the second segment to assume a corner position in which the first segment and the second segment form an angle therebetween of 90 degrees and c) the framing drywall liner is integral across the groove for most of the longitudinal length.

[0017] In another aspect, the drywall apparatus further includes a coating on an external surface of the framing drywall liner for strengthening the groove.

[0018] In a further aspect, the coating comprises a polyester strip such as BoPET (Biaxially-oriented polyethylene terephthalate), that extends over portions of the first and second gypsum-containing segments on either side of the of the groove.

[0019] In an additional aspect, a method of constructing a hingeable drywall apparatus is described, comprising: providing a finishing drywall liner having an internal side and an external side; pouring gypsum slurry onto the internal side of the finishing drywall liner; forming a first groove on a framing drywall liner, the first groove having a shape; placing the framing drywall liner on the gypsum slurry so that the gypsum slurry is sandwiched between the finishing drywall liner and the

framing drywall liner; and allowing the gypsum slurry to set to a hardened gypsum material, wherein, during the step of allowing, the shape of the first groove is impressed upon the slurry resulting in a second groove in the hardened gypsum material that mates with the first groove to thereby form a hinge at the first and second grooves, wherein said first and second grooves give rise to two segments of the apparatus that can rotate about the hinge.

[0020] In yet a further aspect, a drywall system is described comprising: a first gypsum-containing segment having a finishing side, a framing side and a first groove edge; a second gypsum-containing segment having a finishing side, a framing side and a second groove edge; a finishing drywall liner covering the finishing side of the first segment and the finishing side of the second segment to form a hinge therebetween that allows an angle between the first segment and the second segment to vary; a first sheet of framing drywall liner integrally covering the framing side of the first segment and at least a portion of the first groove edge; and a second sheet of framing drywall liner integrally covering the framing side of the second segment and at least a portion of the second groove edge, said first sheet being divided from the second sheet at the hinge, wherein a) the hinge allows the first segment and the second segment to assume a flat position in which the first segment and the second segment are coplanar, b) in the flat position, a groove is formed between the first groove edge and the second groove edge, and c) the hinge allows the first segment and the second segment to assume a corner position in which the first segment and the second segment form an angle therebetween of 90 degrees.

[0021] These together with other aspects and advantages which will be subsequently apparent, reside in the details of construction and operation as more fully hereinafter described and claimed, reference being had to the accompanying drawings forming a part hereof, wherein like numerals refer to like parts throughout.

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## Description

### BRIEF DESCRIPTION OF THE DRAWINGS

[0022] FIG. 1 shows a plan view of a drywall apparatus for forming corners in houses, buildings and the like, according to the principles of the present invention.

[0023] FIGS. 2A and 2B show cross sectional views of a drywall apparatus of FIG. 1.

[0024] FIGS. 2C-2E show various profiles of substantially V-shaped grooves, according to the principles of the present invention.

[0025] FIG. 2F shows finishing paper that is thinner along the groove, according to the principles of the present invention.

[0026] FIG. 2G shows a coating of strengthening material disposed on the internal side of a drywall liner, according to the principles of the present invention.

[0027] FIG. 3 shows a plan view of an outer corner position for drywall apparatus, according to the principles of the present invention.

[0028] FIG. 4 shows a plan view of an inner corner position for the drywall apparatus of FIG. 1.

[0029] FIG. 5 lists steps for constructing a hingeable drywall board composed of gypsum material sandwiched between a finishing drywall liner and a framing drywall liner, according to the principles of the present invention.

[0030] FIG. 6 shows a router for making the drywall apparatus of FIGS. 1-4.

[0031] FIG. 7A shows a flow chart for making a drywall apparatus using a non-subtractive method, according to the principles of the present invention.

[0032] FIG. 7B shows a drywall product resulting from the method outlined in FIG. 7A.

[0033] FIG. 8 shows a first finishing drywall liner placed on a conveyor belt, according to the principles of the present invention.

[0034] FIG. 9A shows a side view of part of an extruder station, according to the principles of the

present invention.

[0035] FIG. **9B** shows a cross section indicated in FIG. **9A** of the extruder station, according to the principles of the present invention.

[0036] FIG. **10** shows a drywall system, according to the principles of the present invention.

[0037] FIG. **11** shows the drywall system of FIG. **10** in an inner corner configuration, according to the principles of the present invention.

[0038] FIG. **12** shows the drywall system of FIG. **10** in one outer corner configuration, according to the principles of the present invention.

[0039] FIG. **13** shows the drywall system of FIG. **10** in another outer corner configuration, according to the principles of the present invention.

[0040] FIGS. **14A** and **14B** show a drywall system, similar to the drywall system of FIG. **10**, in a shipping configuration, according to the principles of the present invention.

[0041] FIG. **15A** shows a drywall system for framing in a flat configuration, according to the principles of the present invention.

[0042] FIG. **15B** shows the drywall system of FIG. **15A** in a framing configuration, according to the principles of the present invention.

[0043] FIG. **15C** shows two drywall systems, each like that shown in FIG. **15B**, mated for producing a window frame, according to the principles of the present invention.

[0044] FIGS. **16A** and **16B** show another drywall system for framing, according to the principles of the present invention.

[0045] FIGS. **17A** and **17B** show another drywall system for framing, according to the principles of the present invention.

[0046] FIGS. **18A** and **18B** show strengthening material applied to the underside of a paper on roll prior to manufacture and extending beyond a largest width on either side of the of the groove.

#### DETAILED DESCRIPTION OF THE PREFERRED EMBODIMENTS

[0047] FIG. **1** shows a plan view and FIGS. **2A** and **2B** show a cross sectional view of a drywall apparatus **10** for forming corners in houses, buildings and the like, according to the principles of the present invention. FIG. **2A** shows the drywall apparatus **10** without a strengthening material **11**, and FIG. **2B** shows the drywall apparatus with the strengthening material **11**, as explained in more detail below. The drywall apparatus **10** includes gypsum material **12**, a finishing drywall liner **14** on a finishing side **16**, and a framing drywall liner **18** on a framing side **20**. The finishing drywall liner **14** has an internal (or slurry) side **22**, which during manufacture of the gypsum board is in contact with gypsum slurry, and an external side **24**, which can be painted, wallpapered, etc.

Likewise, the framing drywall liner **18** has an internal (or slurry) side **26**, which during manufacture of the gypsum board is in contact with gypsum slurry, and an external side **28**, which is in contact with the framing studs or joists when installed.

[0048] The external side **24** of the finishing drywall liner **14** is typically exposed to an observer inside a room. As mentioned above, the external side **24** can be finished with paint or wallpaper, for example. The external side **24** of the framing drywall liner **18** is typically unexposed to an observer inside a room. As mentioned above, the external side **28** of the framing drywall liner **18** faces and is in contact with framing structures, such as wood or metal studs or joists. Sheets of finishing drywall liner **14** may be of a different quality than sheets of framing drywall liner **18**.

Commercially available drywall boards typically consist of gypsum material sandwiched between a finishing drywall liner and a framing drywall liner.

[0049] The gypsum material **12** is divided by a groove **34** into two segments **30** and **32** on either side of the groove **34**. The first gypsum-containing segment **30** is disposed on the internal side **22** of the finishing drywall liner **14**. The second gypsum-containing segment **32** is also disposed on the internal side **22** of the finishing drywall liner **14**. The groove **34** runs between the first segment **30** and the second segment **32**. The finishing drywall liner **14** connects the first segment **30** to the second segment **32**. A hinge axis **36** runs parallel to the groove **34**.

[0050] The first segment **30** and the second segment **32** are capable of hinging about the hinge axis **36**. Moreover, the first segment **30** and the second segment **32** are capable of assuming a flat position in which the first segment **30** and the second segment **32** are coplanar. The first segment **30** and the second segment **32** are shown in the flat position in FIGS. **1**, **2A** and **2B**.

[0051] The finishing drywall liner **14** can have a maximum width **38** at the groove **34** that is at least 40 thousands of an inch when the first gypsum-containing segment **30** and the second gypsum-containing segment **32** are spread apart as much as possible without tearing the drywall liner **14**. In one embodiment, such a width **38** is 52 thousands of an inch. The external side **24** of the finishing drywall liner **14** is divided by the hinge axis **36** into a first external side **40** and a second external side **42**. The groove **34** is formed by forming a first edge **44** having a first edge angle **46** on the first segment **30**, and a second edge **48** having a second edge angle **50** on the second segment **32**. In FIGS. **2A** and **2B**, the first edge angle **46** and the second edge angle **50** are each 45 degrees.

[0052] In FIG. **2A**, the framing drywall liner **18** is disposed to the right and left of the groove **34** but not therein. If desired, the framing drywall liner can line some of the groove. For example, framing drywall liner can span the groove intact. In one embodiment, a tear of the framing drywall liner can be formed in the groove, such as with a saw, scraper or router. The tear divides the framing drywall liner into a first portion and a second portion such that part of the first portion resides in the groove and part of the second portion resides in the groove, wherein the part of the first portion is disposed on the first gypsum-containing segment, and the part of the second portion is disposed on the second gypsum-containing segment.

[0053] The groove **34** in FIGS. **2A** and **2B** is substantially V-shaped in cross section. As used herein, the term “substantially V-shaped” describes not only shapes where the two legs of the V meet at a point, but also blunted shapes in which the two legs of the “V” do not meet at a point, but instead are joined by a flat, such as in FIG. **2A**. The term “substantially V-shaped” also includes shapes where the legs of the V are not perfectly straight, but bowed slightly. FIGS. **2C-E** show various profiles of substantially V-shaped grooves. As explained below, strengthening material can be applied to the groove area. Advantageously, the profiles of FIGS. **2D** and **2E** allow room for the strengthening material and/or drywall liner to reside when the two halves of the drywall apparatus are rotated towards each other, thereby preventing bunching or buckling of strengthening material and/or drywall liner near the hinge. Such bunching or buckling could place unwanted stress on the finishing paper near the hinge resulting in tearing.

[0054] In one embodiment designed to prevent bunching or buckling, and shown in FIG. **2F**, finishing paper **51** is thinner along the groove **53**, such as along and near the hinge axis. For example, the finishing paper **51** could be composed of two plies of paper everywhere except in all or in part of the groove, where it could be one-ply. The one-ply paper would be less likely to bunch up than two-ply paper during hinging because there would be less paper material at the hinge axis. The width of the one-ply paper could be approximately the width of the flat at the groove or somewhat larger or smaller, for example.

[0055] In another embodiment, the finishing paper could be four-ply, except at the groove where it would be three-ply. More generally, according to the principles of the present invention, the finishing paper could be n-ply, except at the groove where it would be m-ply, where  $n > m$ .

[0056] In the embodiment shown in FIGS. **2A** and **2B**, the substantially V-shaped groove has a largest width **52** on the framing side **20** tapering to the narrowest width **38** at the finishing side **16**. For example, the narrowest width **38** can lie in the range of 40 thousands of an inch to 60 thousands of an inch. In FIGS. **2A** and **2B**, showing the drywall apparatus **10** in the flat position, a cross-sectional groove angle **56** subtended by the two legs of the V is 90 degrees.

[0057] Because of the hinging action of the first and second segments, which can damage the finishing drywall liner **14** at the location of the groove **34**, it is preferable to add a strengthening material **11** in the groove. With reference to FIG. **2B**, the strengthening material **11**, such as an elastomer, is applied on at least part of the internal side **22** of the finishing drywall liner **14** at the

groove **34**. The strengthening material **11** is applied for strength or reinforcement of the finishing drywall liner **14** at the groove **34**. The elastomer **11** can include silicone that can be cured by heating or with light, for example. The elastomer can be sprayed on the drywall liner or gypsum. Other strengthening materials include polyester, ethylene-vinyl acetate (EVA), polyurethane and/or acrylic latex. In addition to or instead of an elastomer, a strip of paper, sheet metal or plastic can be applied running along or transverse to the groove **34** to strengthen the finishing drywall liner **14** at the groove **34**. The strip of paper, sheet metal or plastic can be applied with glue, or some other appropriate fastening means. The strengthening material **11** helps prevent the drywall liner **14** connecting the first segment **30** to the second segment **32** from tearing.

[0058] In one embodiment, the strengthening material is an elastomeric coating that is applied at least on part of the internal side of the drywall liner, the elastomeric coating having no other strengthening material applied thereon.

[0059] In another embodiment, as shown in FIGS. **18A** and **18B**, the strengthening material can be a strip **800** of polyester such as BoPET (Biaxially-oriented polyethylene terephthalate), (e.g. Mylar® polyester film manufactured by DuPont-Teijin Corporation), that is applied to the underside of the paper on roll **251** prior to manufacture and extends beyond the largest width **52** on either side of the of the groove **34**. In the event of insufficient crystalline bonding of the strip **800** of polyester to the gypsum material **12** during oven drying, a heat resistant strip or a strip using a resin with a high heat deflection temperature can be included.

[0060] The roll of paper **251** can also include perforations **810**, as shown in FIG. **18A**, through the framing drywall liner **18** on both sides of groove **34** intermediate the edges of the strip **800** and the largest width **52** of the groove **34**. Also shown are additional grooves **815** on either side of the perforations **810**, adapted to mate with corresponding shaped ridges of a structure (not shown) for mounting the framing drywall liner **18** to a wall. Grooves **815** are shown as being rounded with a constant radius, but can be any suitable shape in cross section such as tapered, square, v-shaped, etc. The paper on the roll **251** can be scored with perforations **810** during manufacture so that the strip of paper between the two lines of perforations **810** can be peeled away along the perforations **810** leaving the strip **800** intact and adhered to the gypsum material **12**. In some embodiments, a coating of adhesive can be applied to the gypsum material **12** to bond to the strip **800** to the gypsum material **12** during manufacture.

[0061] With reference to FIG. **2G**, instead or in addition, a coating of strengthening material **41** is completely disposed on the internal side of the drywall liner. In FIG. **2G**, the drywall apparatus is in the flat position, and in such position a) the coating **41** is planar, lying flat on the internal side, and b) at least part of the coating has a dimension **43**, as measured perpendicular to the hinge axis (or the groove, since the hinge axis and the groove are parallel) and parallel to the drywall liner, that is larger than the width **45** at the groove. For example, the strengthening material can be applied on at least part of the internal side **22** of the finishing drywall liner **14** at the groove **34** during the manufacturing process before slurry is poured on the internal side **22**. Advantageously, the strengthening material can prevent the slurry from impregnating the finishing drywall liner at the groove thereby helping to prevent the finishing liner from becoming brittle and cracking when hinged at the groove.

[0062] The drywall apparatus **10** may be used to construct outer or inner corners, in houses, buildings and the like. As used herein, an outer corner is a corner in which an angle  $\theta$  **31** between the first external side **40** of the finishing drywall liner and the second external side **42** of the finishing drywall liner satisfies  $180 \text{ degrees} < \theta < 360 \text{ degrees}$ . Typically, the angle of an outer corner is 270 degrees. As used herein, an inner corner is a corner in which the angle  $\theta$  between the first external side **40** and the second external side **42** satisfies  $0 \text{ degrees} \leq \theta < 180 \text{ degrees}$ . Typically, the angle  $\theta$  of an inner corner is 90 degrees. In the flat position shown in FIG. **2A**, the angle  $\theta$  **31** is 180 degrees. The flat position could be used to construct a flat wall. Thus, advantageously, in addition to forming corners, the drywall apparatus **10** of the present invention can be used to build flat



walls.

[0063] FIG. 3 shows a plan view of an outer corner position for the drywall apparatus 10. Because of the shape of the groove 34 formed from the 45 degree first and second edge angles 46 and 50, the two segments 30 and 32 are capable of hinging about the hinge axis 36 to assume an outer corner position in which the angle between the first external side 40 and the second external side 42 of the finishing drywall liner 14 is 270 degrees. The first edge 44 and the second edge 48 abut at the groove 34 in the outer corner position. This position is suitable for forming a “square” outer corner of a room.

[0064] Advantageously, the width 38 at the groove 34 shown in FIGS. 2A and 2B is there to ensure that the elastomer 11 has a place to reside. In addition, the width 38 yields a slightly rounded corner 39 that is aesthetically pleasing when the angle between the external sides 40 and 42 is 270 degrees for forming an outer corner. The width 38 also relieves stress on the finishing drywall liner 14 so that it does not tear at the groove 34.

[0065] It will be appreciated that the appropriate groove angle is a function of the outer corner angle that one wishes to achieve. Thus, as shown in FIGS. 2A and 2B, to form an exterior angle of 270 degrees, the groove angle 56 of 90 degrees is preferably used as measured in the flat position. In another example, to form an outer corner of 300 degrees, a groove angle of 120 degrees is preferably used. In general, if the angle of the outer corner is x degrees, the groove angle as measured in the flat position is preferably the exterior angle minus 180 degrees.

[0066] FIG. 4 shows a plan view of an inner corner position for the drywall apparatus of FIG. 1. The two segments 30 and 32 are capable of hinging about the hinge axis 36 to assume an inner corner position in which the angle between the first external side 40 and the second external side 42 is 90 degrees.

[0067] It will be appreciated that according to the principles of the present invention, the same drywall apparatus 10 can be used to form an outer and an inner corner.

[0068] FIG. 5 lists steps for constructing a hingeable drywall board composed of gypsum material sandwiched between a finishing drywall liner and a framing drywall liner. The finishing drywall liner may be of a different quality than the framing drywall liner. The external side of the finishing sheet is suitable for finishing the drywall exposed in a room by wallpapering or painting, for example. The framing sheet has an external side for facing and being in contact with the framing structure, such as wood or metal studs.

[0069] Step 100 includes forming a groove on the framing side of the board to a depth reaching the finishing drywall liner. The groove is substantially V-shaped in cross section, as in FIGS. 2A and 2B, with a largest width at the framing side tapering to a narrowest width at the finishing side, wherein the narrowest width is in the range of 30 thousandths of an inch to 60 thousandths of an inch. By forming the groove, a pliable hinge of drywall liner is created. Step 102 includes adding a strengthening material in the groove. For example, an elastomer can be applied at least on the internal side of the finishing liner at the groove. The elastomer strengthens the pliable hinge of drywall liner to prevent tearing of the drywall liner at the groove.

[0070] It should be understood that as used herein drywall board need not refer to just commercially available sizes of boards. Smaller or larger sizes are contemplated. For example, during the manufacturing process, drywall boards that are greater than several hundreds of feet are cut into commercially suitable sizes before being dried in a kiln. The steps listed above can be applied to the board before or after cutting, and before or after drying in the kiln. For example, the strengthening material can be added before or after cutting into commercially suitable sizes, and before or after drying in the kiln.

[0071] One method for making the drywall apparatuses of FIGS. 1-4 involves a router. FIG. 6 shows such a router 100. The router 100 has a frustoconical body 102 and a stem 104. The frustoconical body 102 has a substantially V-shaped cross section with a largest width 106 tapering to a narrowest width 108. The narrowest width is in the range of 30 thousandths of an inch to 60

thousands of an inch.

[0072] Some methods of producing a grooved drywall board capable of hinging may be described as subtractive processes, whereby the groove is formed by removing drywall material from a drywall board, such as by using the router **100**. Instead, as will now be described, a grooved drywall board capable of hinging may be manufactured during the slurry stage by shaping to form a groove as the slurry sets to a hardened gypsum material. Advantageously, waste and gypsum dust, characteristic of a subtractive process, are avoided. In addition, this method affords the opportunity to leave the framing liner intact with or without strengthening material, in contrast to using a router which strips away the framing liner, along with some gypsum material, to form the groove. Time and expense can be saved by forming the groove during the slurry stage. In addition, certain embodiments of the drywall apparatus are best manufactured using a non-subtractive method that does not involve removing gypsum by cutting, scraping or the like. Instead, the drywall board is extruded and formed to have the desired groove that allows the board to hinge.

[0073] FIG. 7A shows a flow chart for making a drywall apparatus using a non-subtractive method, according to the principles of the present invention. In step **200**, a first sheet of finishing drywall liner, having an internal side and an external side, is placed on a conveyor belt external side down. For ease of reference only, the conveyor belt will be assumed to be moving in a north direction. In step **202**, gypsum slurry is poured on the internal side of the first sheet of finishing drywall liner. In step **204**, which can occur before, during or after step **202**, a framing drywall liner is folded to create a first groove. In step **206**, the framing sheet is disposed on top of the slurry with the first groove running substantially in the north-south direction. The resultant slurry sandwich is moved by the conveyor belt to an extrusion station. In step **208**, the slurry sandwich is passed through the extrusion station. Optionally, a roller, or other suitable forming guides, with a substantially V-shaped member complimentary to the first groove of the framing sheet is used at the extrusion station to help set a second groove in the gypsum material that is complimentary to the first groove of the framing sheet. After travelling past the extrusion station, in step **210** the slurry sandwich passes through another roller with a substantially V-shaped member to further help set the second groove in the gypsum material. It will be appreciated that once this setting of the gypsum material occurs, the first groove is nestled in the second groove. In some embodiments, the roller can include ancillary rollers to indent the framing drywall liner **18** with the additional grooves **815**.

[0074] Glue can be applied to the framing paper at the edges before it is placed on top of the finishing paper. The finishing paper is folded to form the edge of the drywall board. For this purpose, the finishing paper will have been pre-creased earlier in the process. Forming guides on the sides of a forming table fold the paper over to shape the edge and the extruder defines the board thickness as it travels through.

[0075] In one embodiment, inline “V-groove” equipment scores or creases three lines in the framing paper to form the V shown in FIG. 2C. The equipment extrudes this shape in the board in the board extruder. V-shaped rollers or continuous guides would also help set the V-groove. To form the substantially V-shaped groove of FIG. 2D, the equipment would score or crease four lines in the framing paper.

[0076] In one embodiment, the final product of the method outlined in FIG. 7A is the drywall apparatus **212** shown in FIG. 7B. The drywall apparatus **212** includes a first sheet of drywall liner **214** having an external side **216** and an internal side **218** opposite the external side **216**. The apparatus also includes a second drywall liner **220**. A first gypsum-containing segment **222** resides between the first sheet **214** and the second sheet **220**. A second gypsum-containing segment **224** also resides between the first sheet **214** and the second sheet **220**. The first segment **222** and second segment **224** are connected by a hinge **226** formed from the first sheet **214** and the second sheet **220** to allow an angle between the first segment and the second segment to vary, such that the hinge allows the first segment and second segment to assume a flat position (shown in FIG. 7B) in which the first segment **222** and the second segment **224** are coplanar (i.e., the first sheet of drywall liner

is substantially planar), wherein, in the flat position, the second sheet **220** includes a substantially V-shaped groove **228** running along the hinge **226**. The groove **228** has a narrower bottom and a wider top, the bottom of the substantially V-shaped groove **228** being within 80 thousands of an inch of the internal side of the first drywall liner. The phrase “within 80 thousands of an inch” includes a preferred embodiment, shown in FIG. 7B, in which the bottom of the substantially V-shaped groove **228** is in contact with the internal side **218** of the first drywall liner **214** (i.e., the second drywall liner **220** is in contact with the first drywall liner **214**).

[0077] In the embodiment shown in FIG. 7B, the second drywall liner **220** is integral along and across the groove **228**. Advantageously, because the second sheet **220** remains integral, the hinge **226** is strengthened. It will be appreciated that in a subtractive process, this feature would be absent. For example, when forming a groove in a gypsum board with the router of FIG. 6, the second sheet **220** would be cut by the router along the gypsum groove, thereby removing framing paper along the groove and reducing strength in that area.

[0078] FIGS. 8, 9A and 9B show some of the components involved in the non-subtractive method of making a drywall apparatus in one embodiment of the present invention.

[0079] In FIG. 8, a first sheet of finishing drywall liner **240**, having an internal side **242** and an external side **244**, is placed on a conveyor belt **246** external side down. Slurry **248** has been poured on the internal side **242**. A framing drywall liner **250** is unspooled from a roll **251** and then disposed on top of the slurry **248** to create a slurry sandwich. In a preferred embodiment, after the framing drywall liner **250** is unspooled from the roll, but before the liner **250** meets the slurry, the liner **250** is folded at a folding station (not shown) to create the appropriate groove on the liner **250**. The folding station can include a crease and/or a scoring saw for this purpose. To form the substantially V-shaped groove of FIG. 7B, for instance, four parallel, longitudinal creased or scored lines have to be produced. The inner two lines are close together to form the bottom flat of the V-shaped groove. Because of their proximity to each other, the two inner lines can be formed by one scoring saw with two adjacent blades, or one W-shaped blade. The shape of the groove will in turn help form a complementary shaped groove in the gypsum when the slurry sets further to the right in FIG. 8.

[0080] In a different embodiment, the paper on the roll **251** is already creased or scored. Pre-creasing or pre-scoring the paper obviates the need to add creasing or scoring machines to the drywall manufacturing line.

[0081] In FIG. 9A, a side view of part of an extrusion station **254** is shown. The slurry sandwich comprising slurry **248** between the finishing drywall liner **240** and the framing drywall liner **250** is passed through the extrusion station **254**. The framing drywall liner **240** has been folded to create a groove (not shown) running from left to right in the FIG. 9A. An extruder member **256** with a substantially V-shaped member **258** is used at the extrusion station **254** to help set a complementary second groove in the gypsum material. The portion of the extrusion station **254** that first engages with the drywall liner **250** on top of slurry (at the left side in FIG. 9A) is slanted to guide and therefore facilitate engagement with the framing drywall liner **250**. In FIG. 9B, the cross section indicated in FIG. 9A of the extruder station **254** is shown.

[0082] In one embodiment, the substantially V-shaped member **258** is as long as the distance travelled by the slurry **248** before it sets. In this manner, during the full setting process from slurry to hardened drywall, there is a force on the framing paper **250**, and on the slurry beneath, to form the substantially V-shaped groove.

[0083] FIG. 10 shows a drywall system **300** consistent with the principles of the present invention. The drywall system **300** includes a first drywall panel **302** covered with a first finishing liner **303**, and a second drywall panel **304** covered with a second finishing liner **305**. The first drywall panel **302** and the second drywall panel **304** are connected to each other via a main hinge **306** along a groove (not shown) that would be behind the page of the figure, like the substantially V-shaped grooves described above. The drywall system **300** also includes a first drywall flap **308** connected

to the first drywall panel **302** via a first hinge **310** along a groove (not shown) that would be behind the page of the figure, like the substantially V-shaped grooves described above, and a second drywall flap **312** connected to the second drywall panel **304** via a second hinge **314** along a groove (not shown) that would be behind the page of the figure, like the substantially V-shaped grooves described above. The first drywall flap **308** is covered with a first flap finishing liner **309**, and the second drywall flap **312** is covered with a second flap finishing liner **313**. FIG. **10** shows the drywall system **300** in a flat configuration in which the panels **302**, **304** and flaps **308**, **312** are all substantially coplanar. In the embodiment shown in FIG. **10**, the first flap **308** has the shape of a right angle triangle with first hypotenuse **315**, and the second flap **312** has the shape of a second right angle triangle with second hypotenuse **317**.

[0084] In the embodiment shown in FIG. **10**, the first finishing liner **303**, the second finishing liner **305**, the first flap finishing liner **309** and the second flap finishing liner **313** are integral across the hinges **306**, **310** and **314** (i.e., no tears across the hinges). Thus, the monikers “first” and “second” in the phrases “first finishing liner” and “second finishing liner,” for example, are not meant to imply that the two liners are non-contiguous pieces with a gap therebetween. Rather, the first finishing liner **303** and the second finishing meet integrally at the hinge **306**, with the hinge **306** demarcating the boundary between the first finishing liner **303** and the second finishing liner **305**. Advantageously, because these finishing liners are integral at the hinges, there is no need to finish the drywall liners at the hinges with drywall tape and compound after the system is affixed to framing members to form a corner in a wall, for example.

[0085] FIG. **11** shows the drywall system **300** of FIG. **10** in an inner corner position or configuration. To arrive at the inner corner configuration, the first drywall panel **302**, the second drywall panel **304**, the first drywall flap **308** and the second drywall flap **312** can hinge so that the first flap finishing liner **309** and the second flap finishing liner **313** lie on a first plane (parallel to the xz plane), the first finishing liner **303** lies on a second plane (parallel to the xy plane) and the second finishing liner **305** lies on a third plane (parallel to the yz plane), such that the first plane, the second plane and the third plane are mutually orthogonal. In the inner corner configuration, the angle between the first finishing liner **303** and the first flap finishing liner **309** is 90 degrees, the angle between the second finishing liner **305** and the second flap finishing liner **313** is 90 degrees, and the angle between the first finishing liner **303** and the second finishing liner **305** is 90 degrees. In the inner corner configuration, the flaps **308** and **312** also mate along their respective hypotenuses **315** and **317**.

[0086] The flaps in FIG. **11**, in the orientation shown, correspond to being on the ceiling. It should be understood that the drywall system can be fastened so that the flaps instead correspond to being on a wall by rotating the configuration appropriately.

[0087] FIG. **12** shows the drywall system **300** of FIG. **10** in a first outer corner configuration. To arrive at the first outer corner configuration, the first drywall panel **302**, the second drywall panel **304**, the first drywall flap **308** and the second drywall flap **312** are rotated starting from the configuration shown in FIG. **10**. In the first outer corner configuration, the angle between the first finishing liner **303** and the first flap finishing liner **309** is 270 degrees, the angle between the second finishing liner **305** and the second flap finishing liner **313** is 270 degrees, and the angle between the first finishing liner **303** and the second finishing liner **305** is 270 degrees. In the first outer corner configuration, the flaps **308** and **312** also mate along their respective hypotenuses **315** and **317**. Again, the first flap finishing liner **309** and the second flap finishing liner **313** lie on a first plane, the first finishing liner **303** lies on a second plane and the second finishing liner **305** lies on a third plane, such that the first plane, the second plane and the third plane are mutually orthogonal. The first outer corner configuration is convenient for building certain bulkheads, for example.

[0088] With reference to FIG. **13**, there is a second outer corner configuration in which the hypotenuses **315** and **317** do not mate. In the second outer corner configuration, the angle between the first finishing liner **303** and the first flap finishing liner **309** is 90 degrees, the angle between the

second finishing liner **305** and the second flap finishing liner **313** is 90 degrees, and the angle between the first finishing liner **303** and the second finishing liner **305** is 270 degrees. The second outer corner configuration is convenient for building an outer corner where two walls meet, for example.

[0089] FIGS. **14A** and **14B** show a drywall system **300a**, which is similar to the drywall system **300** of FIG. **10**, in a shipping configuration; the drywall system **300a** has different dimensions than the drywall system **300**. In particular, the width and length of flaps **308a** and **312a**, corresponding to the two non-hypotenuse sides of each triangle, are of the same length, L. In other words, in plan view, each flap has the shape of an isosceles, right angle triangle with each of the two equal sides having a length L. Also, in plan view, each panel has the shape of a square with sides equal to L. FIG. **14A** shows the drywall system **300a** that has been folded into a shipping configuration, leaving a void **320**. In the shipping configuration, the angle between the first finishing liner and the first flap finishing liner is zero degrees, the angle between the second finishing liner and second flap finishing liner is also zero degrees, and the angle between the first finishing drywall liner and the second finishing drywall liner is 180 degrees. In other words, in the shipping configuration, the finishing liners of the first drywall panel and the first drywall flap face and are in contact with each other; likewise, in the shipping configuration, the finishing liners of the second drywall panel and the second drywall flap face and are in contact with each other. In the shipping configuration, the first drywall panel and the second drywall panel lie flat.

[0090] FIG. **14B** shows the same system **300a** with a cardboard insert **322** filling the void to provide rigidity to the system **300a** thereby helping to prevent tearing of the flaps **308a** and **312a**. To this end, tape **324** is also applied to the system **300a** around edges. The result is a rigid system that is easy to transport and less likely to tear.

[0091] In another possible shipping configuration, consistent with the principles of the present invention, the angle between the first finishing liner and the first flap finishing liner is 180 degrees, the angle between the second finishing liner and second flap finishing liner is also 180 degrees, and the angle between the first finishing drywall liner and the second finishing drywall liner is zero degrees.

[0092] FIG. **15A** shows another drywall system **500** for framing a window, consistent with the principles of the present invention. The system **500** includes a first drywall panel **502** having a first finishing drywall liner **503**, and a second drywall panel **504** having a second finishing drywall liner **505**, the second drywall panel **504** connected to the first drywall panel **502** via a main hinge **506**. The hinge **506** can be formed by forming a groove (not shown) on what corresponds to the back of the page of the figure. The groove can be like the substantially V-shaped grooves described above. A drywall flap **508**, having a flap finishing liner **509**, is connected to the first drywall panel **502** via a flap hinge **510**. In the embodiment shown in FIG. **15A**, the bottom of the drywall system has a forty-five degree straight cut to mate with a second drywall system to form a window frame, as detailed below.

[0093] The first drywall panel **502** and the second drywall panel **504** can hinge about the main hinge **506** so that the system can assume a window frame configuration in which the first finishing liner **503** lies on a first plane, b) the second finishing liner **505** lies on a second plane and c) the flap finishing liner **509** lies on a third plane, such that the first plane, the second plane and the third plane are mutually orthogonal. Starting from the system in the substantially flat position shown in FIG. **15**, one can arrive at the window frame configuration by rotating the second drywall panel **504** ninety degrees about the main hinge **506** towards the back of the page. Next, the flap **508** is rotated, towards the front of the page, ninety degrees about the flap hinge **510**. Thus, in the window frame configuration, the angle between the first finishing liner **503** and the second finishing liner **505** is 270 degrees, and the angle between the first finishing liner **503** and the flap finishing liner **509** is 90 degrees. The resultant window frame configuration is shown in FIG. **15B**.

[0094] FIG. **15C** shows the system **500** depicted in FIG. **15B**. In addition, a similar second system

**520** is also shown, also in a window frame configuration. The two systems **500** and **520** are shown mated together to form one corner of a window frame. To form a full window frame, four such systems are required to construct four corners. It will be appreciated that in the window frame configuration shown in the embodiment of FIG. **15C**, the first drywall panel **502** and the flap **508** are forming an inner corner, and the first drywall panel **502** and the second drywall panel **504** are forming an outer corner, as these terms are defined above.

[0095] Advantageously, the seams formed between the two systems lie on a plane, instead of at the intersection of two planes, thus making it easier to finish the seams by taping, applying joint compound and sanding. Also advantageously, to form a square window frame, four identical systems **500** can be used, thus reducing the number of different components required to build such a frame.

[0096] The inventor contemplates several modifications to or embodiments of the system **500** shown in FIGS. **15A-C**. First, as mentioned in the last paragraph, it will be appreciated that if four substantially identical systems like **500** are used, the resultant window frame will be square. For a rectangular, non-square window frame, a first pair of identical systems and a second pair of identical systems can be used, such that the second pair is longer than the first pair. In such case, a rectangular, non-square window frame will result.

[0097] Second, the system **500** makes use of forty-five degree angles. For example, the flap **508** describes an isosceles, right angled triangle, so that the angle between the hypotenuse and the hinge **510** is forty-five degrees, and correspondingly, an angle referenced as **511** in FIG. **15A** is also forty-five degrees. It will be appreciated that other angles can be used. For example, the aforementioned angle between the hypotenuse and the hinge **510** can be larger than forty-five degrees provided the angle **511** is correspondingly smaller than forty-five degrees (or vice versa) so that two systems can mate to form a window frame when in the window frame configuration. Specifically, the sum of these two angles should be ninety degrees. Same considerations apply to the angles on the other panel **504**. In principle, the angle **511** can approach ninety-degrees, but if ninety degrees is used (implying that the angle between the hypotenuse and the hinge **510** is zero degrees), it will be appreciated that the seam formed between the two systems will lie at the intersection of two planes, which is less desirable.

[0098] Third, the system **500** can be modified to produce two different corner systems, which together with a hinging rectangular system similar to the one shown in FIG. **1**, can be used to build a window frame. Specifically, with reference to the orientation of system **500** shown in FIG. **15A**, one of the two different corner systems would be obtained from system **500** by cutting a bottom portion of the system **500** to leave a horizontal bottom edge (i.e., an edge perpendicular to the left and right sides of the system **500** shown in FIG. **15A**); the other one of the two different corner systems would be obtained from system **500** by cutting a top portion of the system **500** shown in FIG. **15A** to leave a horizontal top edge (i.e., perpendicular to the left and right sides of the system **500** shown in FIG. **15A**). Corners of the frame can be constructed by mating two such different corner systems. Middle (non-corner) sections of the frame can be constructed from a system similar to FIG. **1** by abutting an end to the aforementioned edges.

[0099] In some of the embodiments described above, a cove or depression may be added near edges that form seams. The term “seam” refers to a region where two drywall boards abut. Seams typically have to be finished by adding drywall tape and compound, followed by sanding. The cove or depression helps in this finishing process by acting as a reservoir for the compound. For example, in FIG. **16A**, a drywall system **600** is shown for forming a window frame. The drywall system **600** is similar to the drywall system **500**, except that depressions **602** are shown along what will form seams when abutted to other drywall systems. FIG. **16B** shows a cross-sectional view as indicated in FIG. **16A**, which is similar to the cross-sectional view of FIG. **2A**. In other systems, these depressions are useful on the finishing side along any seam that will need finishing, such as along the hypotenuse **315** and hypotenuse **317** of the system **300** shown in FIG. **10**.

[0100] FIGS. 17A and 17B show another drywall system **700** suitable for framing, such as window framing. Several of the features of the drywall system **700** are similar to the drywall system **500** of FIG. 15A, but one difference is that the drywall system **700** has no flap. The drywall system **700** includes a first drywall panel **702** having a first finishing drywall liner **703**. The first drywall panel **702** is hingeably connected to a second drywall panel **704** having a second finishing drywall liner **705**. The first drywall panel **702** and the second drywall panel **704** are connected at a hinge **706**, and are capable of hinging about a hinge axis **707**. As mentioned above with respect to FIG. 10, the monikers “first finishing drywall liner” and “second finishing drywall liner” denote two regions continuously connected across their boundary (the hinge **706**), similar to FIG. 2A or 2B. On the opposite side than the one shown in FIG. 17A, there is a groove (not shown) like that appearing in FIG. 2A or B, which will not be described here again. The groove permits the first drywall panel and the second drywall panel to assume an angle therebetween greater than 180 degrees, where, again, the angle (corresponding to **6** in FIG. 2A) is measured between the first finishing drywall liner **703** and the second finishing drywall liner **705** on the side of the liners not containing gypsum material.

[0101] As shown in FIGS. 17A and 17B, the second drywall panel **704** describes a trapezoid with only two sides **708**, **709** parallel. By hinging the system **700** so that the angle between the first drywall panel **702** and the second drywall panel **704** is 270 degrees (“frame configuration”), a window frame can be constructed, as shown in Figure

[0102] It should be understood that in the following method claims, the order in which the steps are listed need not correspond to the temporal order in which the steps are taken in practice. In particular, in the method of constructing a hingeable drywall apparatus, the step of forming a first groove on a framing drywall liner may be performed after the step of placing the framing drywall liner on the gypsum slurry. For example, the first groove on the framing drywall liner may be formed at the same time that the shape of the first groove is impressed upon the slurry, resulting in the second groove in the hardened gypsum material.

[0103] The many features and advantages of the invention are apparent from the detailed specification and, thus, it is intended by the appended claims to cover all such features and advantages of the invention that fall within the true spirit and scope of the invention. Further, since numerous modifications and changes will readily occur to those skilled in the art, it is not desired to limit the invention to the exact construction and operation illustrated and described, and accordingly all suitable modifications and equivalents may be resorted to, falling within the scope of the invention.

## Claims

**1-34.** (canceled)

**35.** A drywall apparatus comprising: a finishing drywall liner; a framing drywall liner; a first gypsum-containing segment between the finishing drywall liner and the framing drywall liner; and a second gypsum-containing segment between the finishing drywall liner and the framing drywall liner, said first segment and second segment connected by a hinge formed from the finishing drywall liner and the framing drywall liner to allow an angle between the first segment and the second segment to vary, such that the hinge allows the first segment and the second segment to assume a flat position in which the first segment and the second segment are coplanar, wherein a) in the flat position, the framing drywall liner includes a groove having a longitudinal length running along the hinge, b) the hinge allows the first segment and the second segment to assume a corner position in which the first segment and the second segment form an angle therebetween of 90 degrees and c) the framing drywall liner is integral across the groove for most of the longitudinal length, wherein the hinge is configured such that the first segment and the second segment are capable of assuming: a) an inner corner position in which the angle of 90 degrees is

measured between the finishing drywall liner on one side of the hinge and the finishing drywall liner on another side of the hinge; and b) an outer corner position in which the angle of 90 degrees is measured between the framing drywall liner on one side of the hinge and the framing drywall liner on another side of the hinge.

**36.** The drywall apparatus of claim 35, wherein a) the first segment has a first edge on one side of the hinge, said first edge being parallel and distal to the hinge, and the second segment has a second edge on an other side of the hinge, said second edge being parallel and distal to the hinge, b) the first segment has a framing side that stretches from the groove to the first edge and the second segment has a framing side that stretches from the groove to the second edge, and c) the framing drywall liner integrally covers an area that includes most of the framing side of the first segment and most of the framing side of the second segment.

**37.** The drywall apparatus of claim 35, such that the first gypsum-containing segment includes a first angled edge located at the groove and the second gypsum-containing segment includes a second angled edge located at the groove, the first angled edge and the second angled edge forming a V-shape, wherein the first angled edge and the second angled edge is in contact with the finishing drywall liner.

**38.** The drywall apparatus of claim 35, such that the first gypsum-containing segment includes a first angled edge located at the groove and the second gypsum-containing segment includes a second angled edge located at the groove, the first angled edge and the second angled edge forming a V-shape, wherein the first angled edge and the second angled edge have some gypsum that impregnates the finishing drywall liner due to gypsum slurry having come in contact therewith during a manufacturing stage.

**39.** The drywall apparatus of claim 35, wherein the groove is formed using a non-subtractive method.

**40.** The drywall apparatus of claim 35, wherein the groove is formed by allowing a gypsum slurry to set to a groove shape.

**41.** The drywall apparatus of claim 35, wherein the groove is formed by extrusion.

**42.** The drywall apparatus of claim 35, further including additional grooves on either side of the grooves adapted to mate with a structure for mounting the framing drywall liner to a wall.

**43.** The drywall apparatus of claim 35, further including a coating on an external surface of the framing drywall liner for strengthening the groove.

**44.** The drywall apparatus of claim 43, wherein the coating comprises a polyester strip that extends over portions of the first and second gypsum-containing segments on either side of the groove.

**45.** The drywall apparatus of claim 44, wherein the polyester strip comprises BoPET (Biaxially-oriented polyethylene terephthalate).

**46.** The drywall apparatus of claim 44, wherein the polyester strip is heat resistant.

**47.** The drywall apparatus of claim 44, wherein the polyester strip is coated with a resin having a high heat deflection temperature.

**48.** The drywall apparatus of claim 44, further including a coating of adhesive applied to the first and second gypsum-containing segments for bonding the polyester to the first and second gypsum-containing segments.

**49.** The drywall apparatus of claim 35, further comprising a coating on at least part of an internal surface of the finishing drywall liner for strengthening the groove.

**50.** A drywall system comprising: a first gypsum-containing segment having a finishing side, a framing side and a first groove edge; a second gypsum-containing segment having a finishing side, a framing side and a second groove edge; a finishing drywall liner covering the finishing side of the first segment and the finishing side of the second segment to form a hinge therebetween that allows an angle between the first segment and the second segment to vary; a first sheet of framing drywall liner integrally covering the framing side of the first segment and at least a portion of the first groove edge; and a second sheet of framing drywall liner integrally covering the framing side of the



second segment and at least a portion of the second groove edge, said first sheet being divided from the second sheet at the hinge, wherein a) the hinge allows the first segment and the second segment to assume a flat position in which the first segment and the second segment are coplanar, b) in the flat position, a groove is formed between the first groove edge and the second groove edge, and c) the hinge allows the first segment and the second segment to assume a corner position in which the first segment and the second segment form an angle therebetween of 90 degrees, wherein the hinge is configured such that the first segment and the second segment are capable of assuming: a) an inner corner position in which the angle of 90 degrees is measured between the finishing drywall liner on one side of the hinge and the finishing drywall liner on another side of the hinge; and b) an outer corner position in which the angle of 90 degrees is measured between the framing drywall liner on one side of the hinge and the framing drywall liner on another side of the hinge.

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